Hall Ticket Number:

MR-21

Code: B415CS5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, July 2025

Computer Organization and Architecture

(Computer Science and Engineering)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What are the basic functional units of a computer?	2	1	1	1
1. b)	Explain arithmetic, logical and shift operations with examples.	2	3	1	1
1. c)	Write the differences between Hardwired Control and micro programmed control.	2	1	2	2
1. d)	List out the types of addressing modes.	2	3	2	2
1. e)	Elaborate on floating point-addition and subtraction.	2	3	3	2
1. f)	Do the subtraction using complements 100-11100.	2	4	3	2
1. g)	Explain the types of modes of transfer with an example.	2	1	4	3
1. h)	List out the types of memory techniques.	2	3	4	3
1. i)	Compare RISC & CISC characteristics.	2	3	5	2
1. j)	Discuss the difference between tightly coupled multiprocessor and loosely coupled multiprocessor.	2	2	5	2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	With the help of block diagram, explain common bus system.	5	1	1	1
2. b)	In the basic computer each instruction cycle consists of four phases. List and explain them.	5	4	1	1
	OR				
2. c)	Design a 4 bit Arithmetic Circuit.	5	4	1	1
2. d)	Explain role of different computer registers with respect to instruction Fetch cycle and execution. Give example	5	1	1	1
3. a)	What is control memory? Write about address sequencing.	5	3	2	2
3. b)	Explain direct addressing mode and indirect addressing mode with an example.	5	1	2	2
	OR				
3. c)	With the diagram explain the basic organization of micro programmed control unit.	5	1	2	2
3. d)	Give the instruction format of different instruction of computing system.	5	2	2	2
4. a)	Briefly discuss the hardware implementation of Booth algorithm.	5	2	3	3
4. b)	Explain fixed point addition and subtraction in computer arithmetics with an example.	5	4	3	3
	OR				
4. c)	Perform the following using Booth algorithm 13 x -5.	5	4	3	3
4. d)	Write a short note on floating point addition of numbers with an example.	5	2	3	3
5. a)	Explain the memory hierarchy with a neat diagram.	5	1	4	2
5. b)	With the neat diagram explain the concepts of direct memory access.	5	2	4	2

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	OR				
5. c)	List and explain different types of cache memory mapping techniques.	5	3	4	3
5. d)	Differentiate between RAM and ROM.	5	2	4	3
6. a)	With the diagram explain the crossbar switch interconnection structure for 8 CPUs and its memory module.	5	2	5	2
6. b)	With the neat diagram explain the concepts of pipelining with an example.	5	1	5	2
	OR				
6. c)	Illustrate multistage switching network with an example.	5	1	5	2
6. d)	Explain the concept of cache coherence problem and its solution.	5	2	5	2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome